

Documentation Index — Neuromorphic Sleep Stage Scoring

Documentation Structure

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├─ architecture.md      # Detailed model architecture
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├─ methodology.md       # Research methodology
├─ results.md           # Official final results
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Documentation by Audience

For Researchers

Document	Description
architecture.md	Model design decisions and parameter analysis
methodology.md	Research approach and experimental design
dataset.md	Data sources, preprocessing, and contracts
results.md	Official metrics and interpretation

For Engineers

Document	Description
architecture.md	Input/output specifications and components
deployment.md	Export, quantization, and edge deployment
dataset.md	Data formats and cache structure
methodology.md	Training configuration and reproducibility

For Exhibition

Document	Description
exhibition.md	Demo script, poster layout, Q&A
results.md	Official result block for display
team.md	Team roles for poster/presentation
architecture.md	Visual architecture diagrams

Quick Reference

Official Result (15-Subject Expanded)

Improved Student
 87.5% $\pm$ 3.2% Test Accuracy
 Cohen's κ = 0.763 $\pm$ 0.043
 Macro F1 = 0.721 $\pm$ 0.050
 99,477 Parameters
 8.5 ms/batch CPU latency

Historical Result (4-Subject Baseline)

Improved Student
 87.34% Test Accuracy
 Cohen's κ = 0.7551
 99,477 Parameters

LoRA Adaptation Result

LoRA $r=8$
 90.66% $\pm$ 3.59% Held-Out-Subject CV Test Accuracy
 Cohen's κ = 0.8092 $\pm$ 0.0663
 552 Trainable Parameters (0.55%)

Model Checkpoint

artifacts/student_improved_best.pt

Run Commands

```
# Verify model
python scripts/verify_model.py

# LoRA cross-validation
python scripts/run_lora_cv.py

# Train LoRA adapter
python scripts/train_lora_adapter.py --rank 8 --alpha 16 --output artifacts/lora/head_r8

# Evaluate LoRA adapter
python scripts/evaluate_lora_adapter.py --adapter artifacts/lora/head_r8

# Dashboard
streamlit run app/streamlit_app.py

# Tests
python -m pytest tests/ -v
```

Team Contact

Member	Role
Param Kaushik	Dataset & Data Governance
Suha Vora	Signal Preprocessing
Shailendra Bhatt	Exploratory Data Analysis
Shamique Khan	Model Development & Training
Aasir Jaffer Lone	Evaluation & Performance

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